İstanbul Sabahattin Zaim University

BIM 439 DATA MINING

Screw Head Detection Project

Group No: 3

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Contents

1 Introduction

This project focuses on the detection and classification of screw heads using the YOLOv8 object detection model

2 Literature Review

For the literature review, we used YOLO offical website and lecture slides.

3 Framework Structure, Components and Architecture

We utilized the YOLOv8 Medium model to achieve higher accuracy. To convert the dataset into the YOLO format, we used Roboflow.

4 Software Usage

- Step 1 → Find data
- Step 2 → Label data and change it to YOLO format
- Step 3 → Use YOLO and train
- Step 4 → Use Model for test

5 Algorithm Description of the Framework

We used YOLOv8, a CNN with complex algorithms, to perform screw head classification and object detection.

6 Runtime Samples

We used Roboflow to convert the dataset. The raw annotated images were uploaded to the platform, which facilitated the automatic conversion of annotations into the YOLOv8-compatible format.

7 Results and Discussion

We have 84% mAP@50 and you can look at Figure 3 for scores

```
  Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 54/100       6.6G     0.7911     0.4505     0.8297         41        640: 100% ━━━━━━━━━━ 423/423 1.8it/s 4:00<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━ 10/10 2.3it/s 4.4s.5ss
                   all        301       1725      0.973      0.946      0.979      0.817

  Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 55/100      6.53G     0.7942     0.4498     0.8314         31        640: 100% ━━━━━━━━━━ 423/423 1.8it/s 4:00<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━ 10/10 2.3it/s 4.4s.5ss
                   all        301       1725      0.975      0.959      0.983      0.824

  Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 56/100      6.51G     0.7864     0.4439     0.8307         63        640: 100% ━━━━━━━━━━ 423/423 1.8it/s 4:00<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━ 10/10 2.3it/s 4.4s.5ss
                   all        301       1725      0.965      0.964      0.983      0.821

  Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 57/100       6.6G     0.7849     0.4408      0.829         36        640: 100% ━━━━━━━━━━ 423/423 1.8it/s 3:60<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━ 10/10 2.3it/s 4.4s.5ss
                   all        301       1725      0.966      0.957      0.983      0.822
```

Figure 1: mAP@50

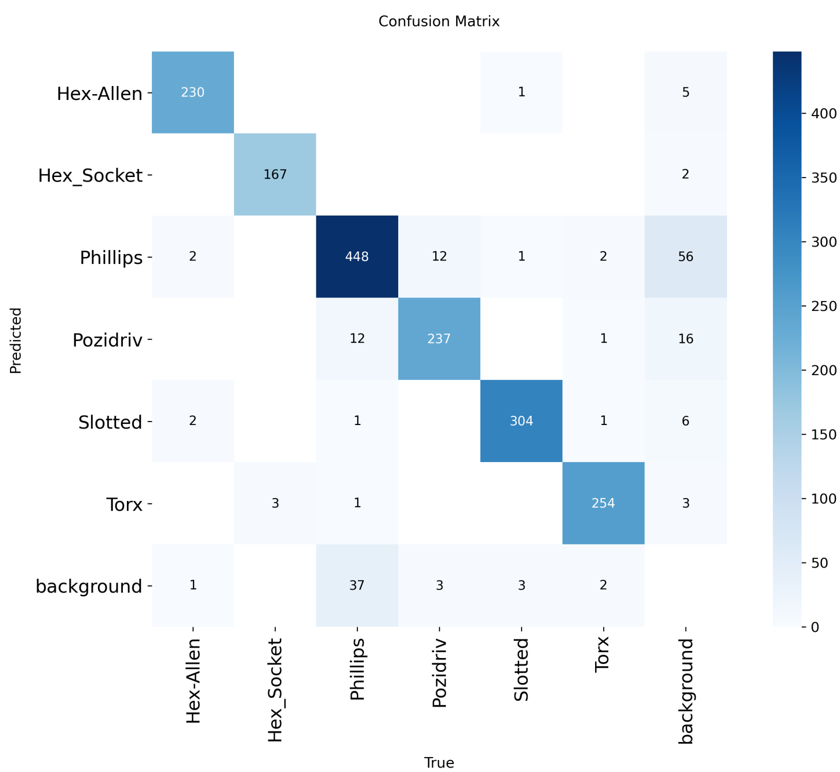

Figure 2: Confusion Matrix

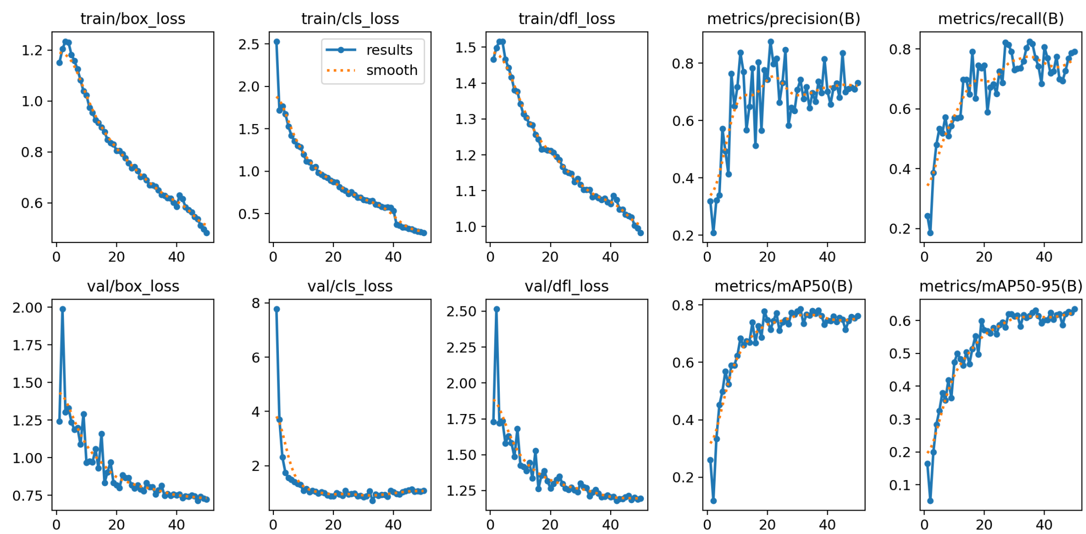

Figure 3: All Scores

8 Future Work

In future works, we plan to add more screw types and calculate the length of the screws.

9 References

- https://docs.ultralytics.com/
- https://pjreddie.com/
- https://roboflow.com/